In [1]:
```
pip install nltk
```

```
Requirement already satisfied: nltk in c:\users\bhagy\appdata\local\programs\python
\python311\lib\site-packages (3.9.4)
Requirement already satisfied: click in c:\users\bhagy\appdata\local\programs\python
\python311\lib\site-packages (from nltk) (8.3.2)
Requirement already satisfied: joblib in c:\users\bhagy\appdata\local\programs\pytho
n\python311\lib\site-packages (from nltk) (1.5.3)
Requirement already satisfied: regex>=2021.8.3 in c:\users\bhagy\appdata\local\progr
ams\python\python311\lib\site-packages (from nltk) (2026.4.4)
Requirement already satisfied: tqdm in c:\users\bhagy\appdata\local\programs\python
\python311\lib\site-packages (from nltk) (4.67.3)
Requirement already satisfied: colorama in c:\users\bhagy\appdata\local\programs\pyt
hon\python311\lib\site-packages (from click->nltk) (0.4.6)
Note: you may need to restart the kernel to use updated packages.
[notice] A new release of pip is available: 24.0 -> 26.2.1
[notice] To update, run: python.exe -m pip install --upgrade pip
```

In [2]:
```
import nltk
from nltk.corpus import stopwords
from nltk.tokenize import word_tokenize
from nltk.stem import PorterStemmer

nltk.download('punkt')
nltk.download('punkt_tab')
nltk.download('stopwords')

print("Libraries imported successfully.")
```

```
[nltk_data] Downloading package punkt to
[nltk_data]     C:\Users\bhagy\AppData\Roaming\nltk_data...
[nltk_data]   Package punkt is already up-to-date!
[nltk_data] Downloading package punkt_tab to
[nltk_data]     C:\Users\bhagy\AppData\Roaming\nltk_data...
[nltk_data]   Package punkt_tab is already up-to-date!
[nltk_data] Downloading package stopwords to
[nltk_data]     C:\Users\bhagy\AppData\Roaming\nltk_data...
Libraries imported successfully.
[nltk_data]   Package stopwords is already up-to-date!
```

In [3]:
```
text = """
The students are learning Python programming.
They are studying different programming languages.
The students were working on their projects.
"""

print("Original text:")
print(text)
```

```
Original text:

The students are learning Python programming.
They are studying different programming languages.
The students were working on their projects.
```

In [4]:
```python
words = word_tokenize(text)

print("Tokens:")
print(words)
```

```
Tokens:
['The', 'students', 'are', 'learning', 'Python', 'programming', '.', 'They', 'are',
'studying', 'different', 'programming', 'languages', '.', 'The', 'students', 'were',
'working', 'on', 'their', 'projects', '.']
```

In [5]:
```python
stop_words = set(stopwords.words('english'))

filtered_words = [
    word for word in words
    if word.lower() not in stop_words
]

print("Text after stop-word removal:")
print(filtered_words)
```

```
Text after stop-word removal:
['students', 'learning', 'Python', 'programming', '.', 'studying', 'different', 'pro
gramming', 'languages', '.', 'students', 'working', 'projects', '.']
```

In [6]:
```python
stemmer = PorterStemmer()

stemmed_words = [
    stemmer.stem(word) for word in filtered_words
]

print("Text after stemming:")
print(stemmed_words)
```

```
Text after stemming:
['student', 'learn', 'python', 'program', '.', 'studi', 'differ', 'program', 'langua
g', '.', 'student', 'work', 'project', '.']
```

In [7]:
```python
processed_text = " ".join(stemmed_words)

print("Final processed text:")
print(processed_text)

with open("filteredtext.txt", "w", encoding="utf-8") as file:
    file.write(processed_text)

print("Processed text saved to filteredtext.txt")
```

```
Final processed text:
student learn python program . studi differ program languag . student work project .
Processed text saved to filteredtext.txt
```